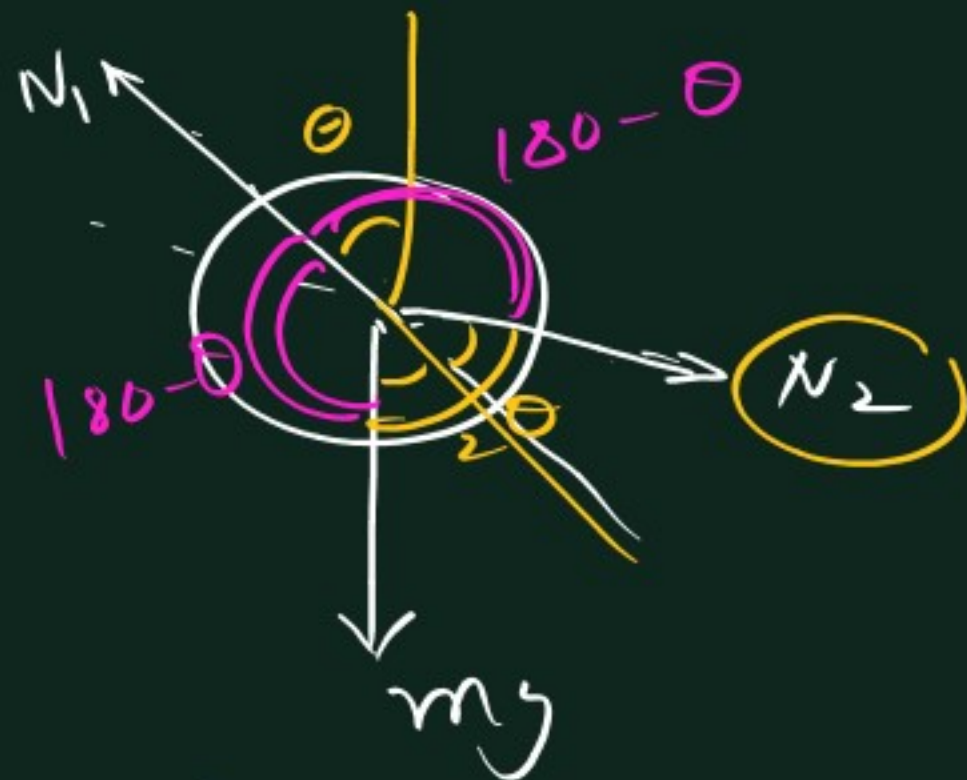
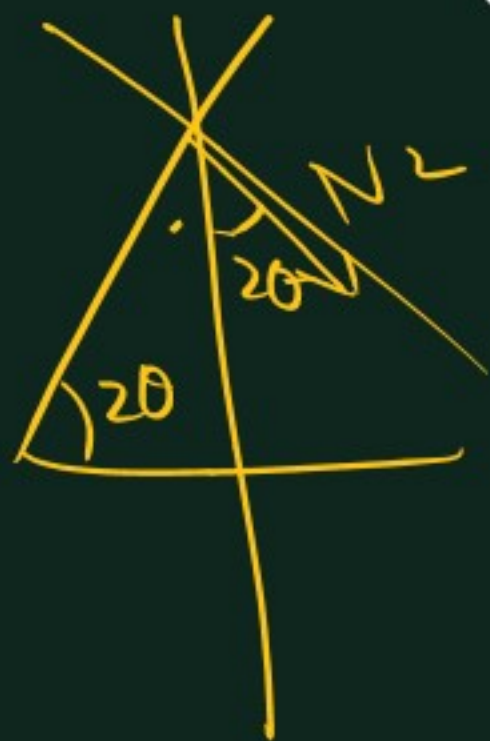
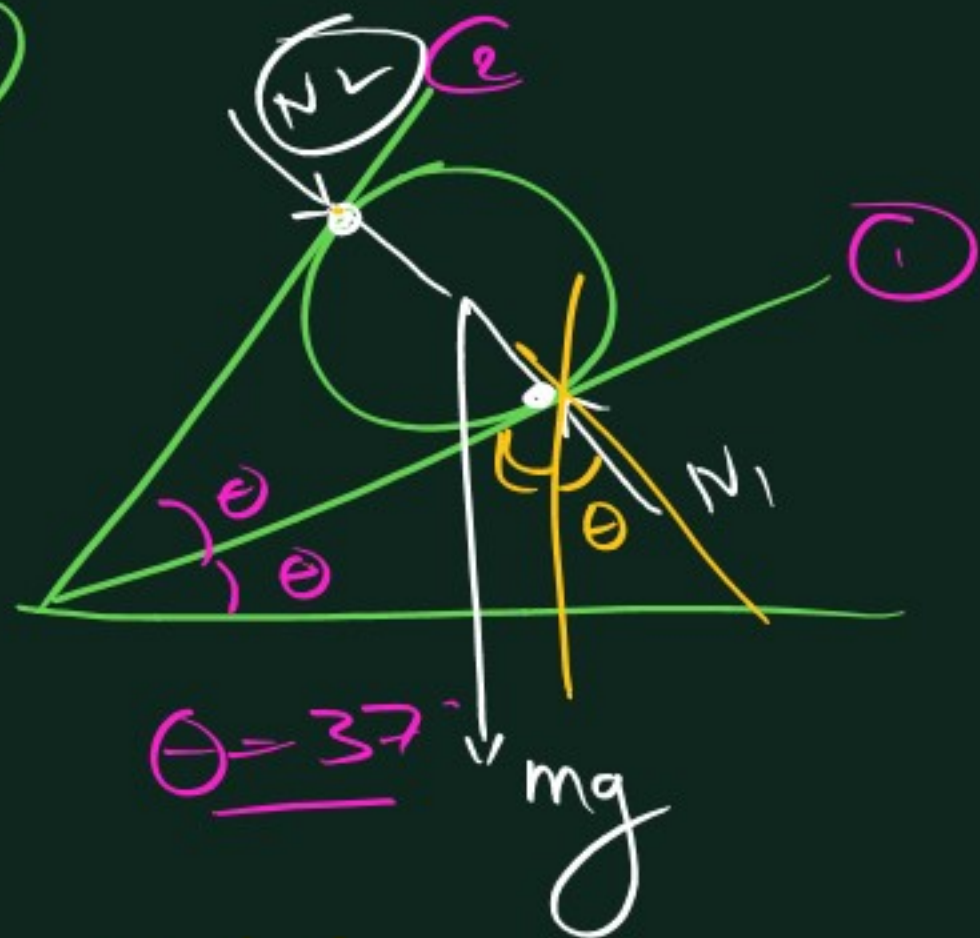


$\frac{L+L}{5}$

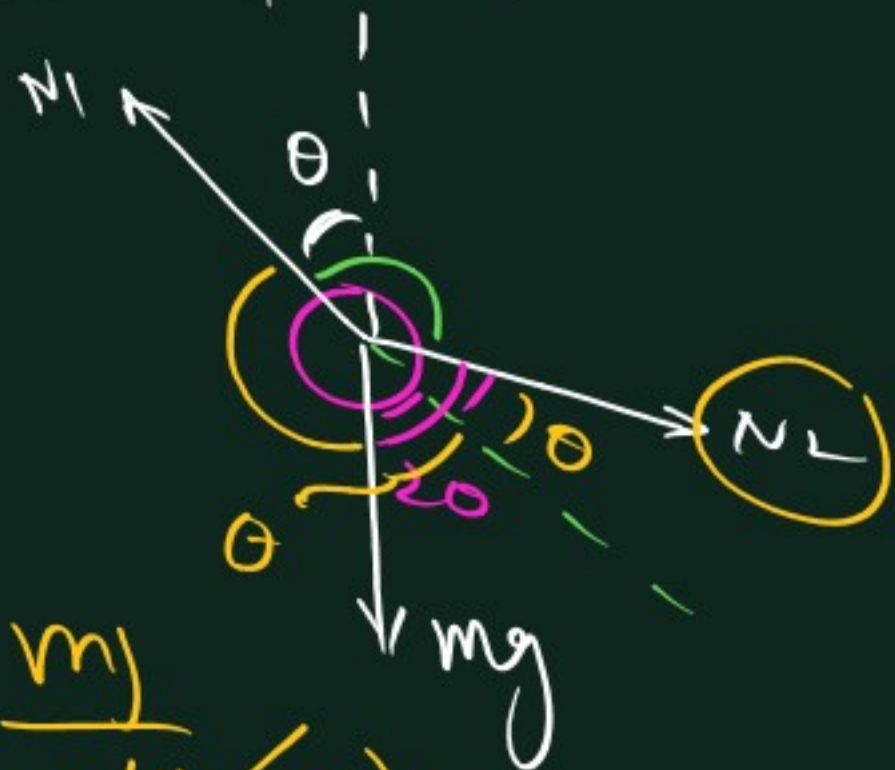
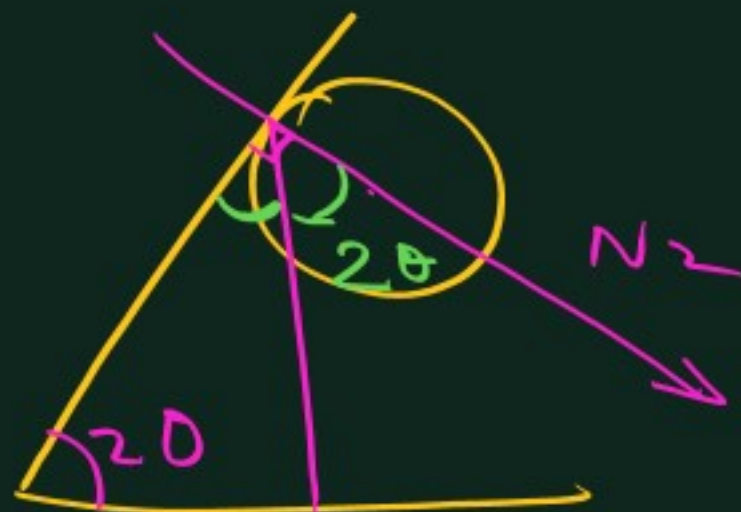
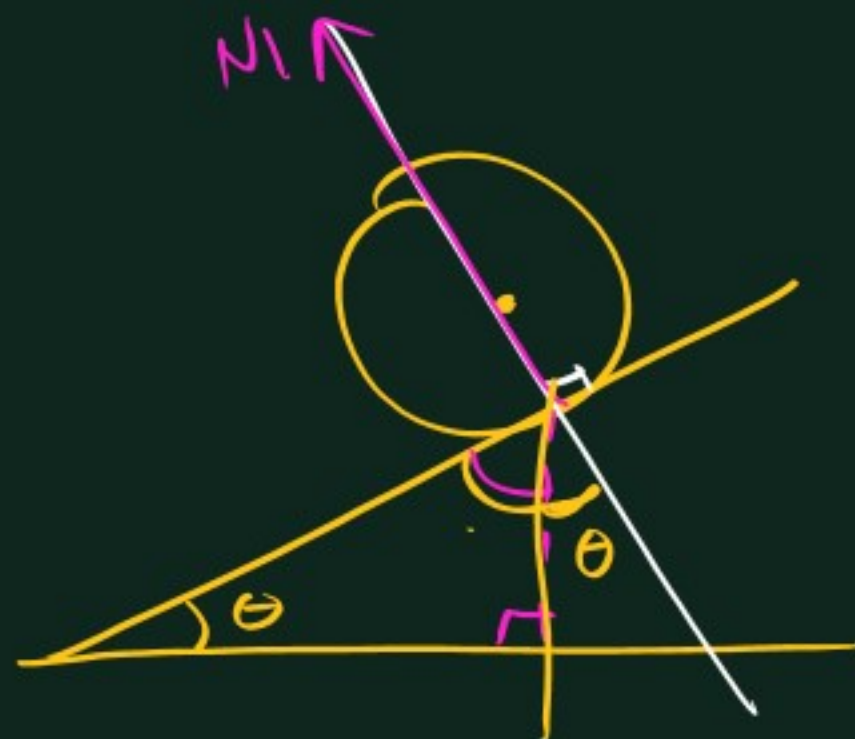
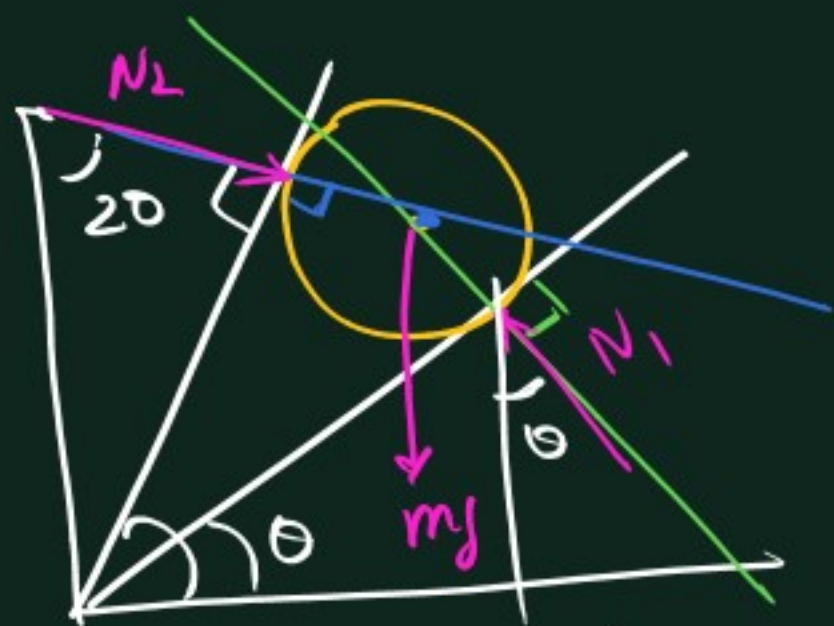


$$\sin(80 - \theta) = \sin \theta$$

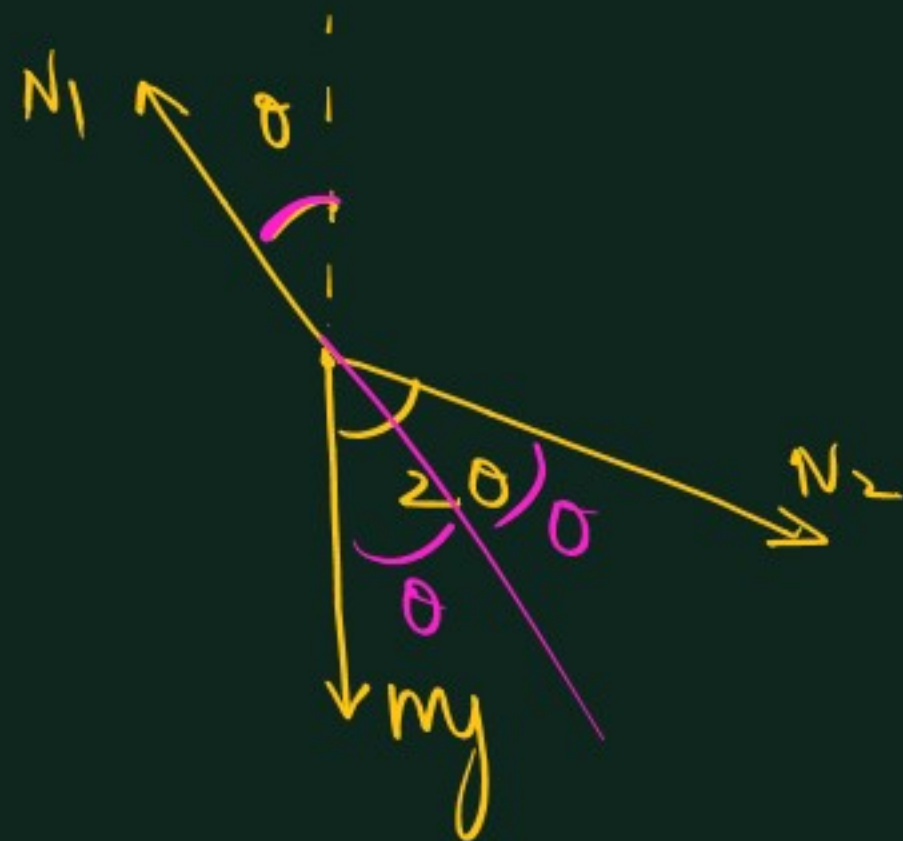
Laminar theorem

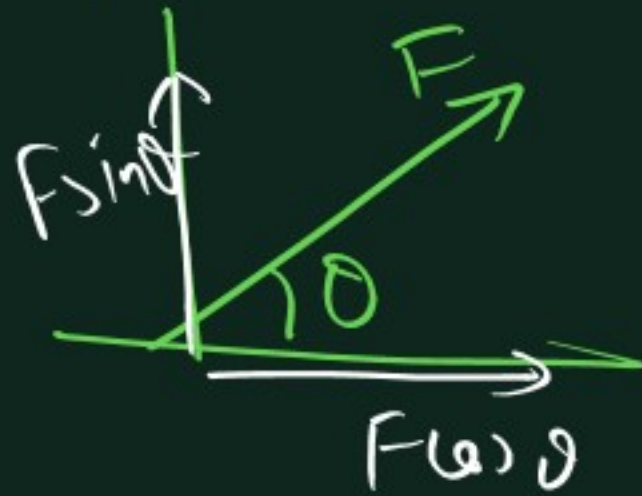
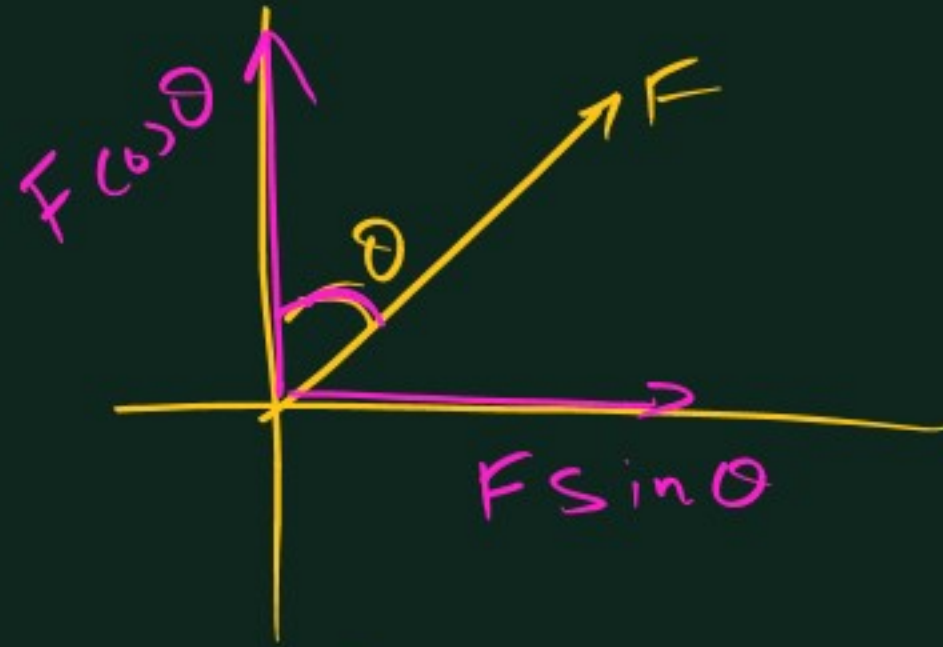
$$\frac{N_2}{\sin(180 - \theta)} = \frac{mg}{\sin(180 - \theta)} = \frac{N_1}{\sin 2\theta}$$

$$\Rightarrow \boxed{N_2 = mg}, \quad N_1 = \frac{\sin 2\theta}{\sin \theta} mg = 2mg \cos \theta$$



$$\frac{N_2}{\sin(180^\circ - \theta)} = \frac{mg}{\sin(180^\circ - \theta)}$$





$$kx = mg$$

$$\Rightarrow x = \frac{mg}{k}$$

Acceleration based problems

Step I → FBD & components

along a ⊥ to a

Step II Along a

$$\Sigma F = ma$$

⊥ to a

$$\Sigma F = 0$$

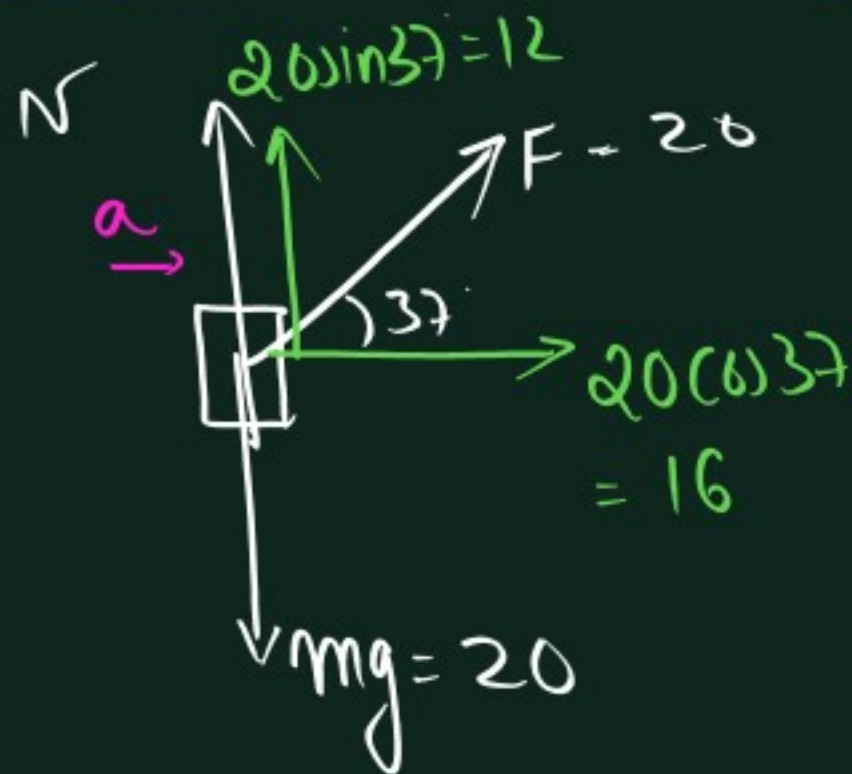
Q1



Find 1) a
2) N

$$\sin 37^\circ = 3/5$$

$$\cos 37^\circ = 4/5$$



Along a

$$\Sigma F = ma$$

$$16 = 2 \times a$$

$$\Rightarrow \boxed{a = 8}$$

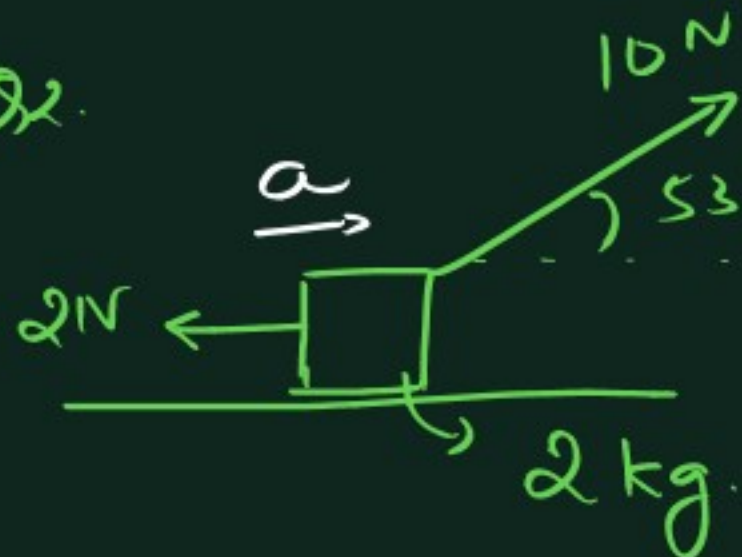
⊥ to a, $\Sigma F = 0$

Up Force = Down Force

$$\Rightarrow N + 12 = 20$$

$$\Rightarrow \boxed{N = 8}$$

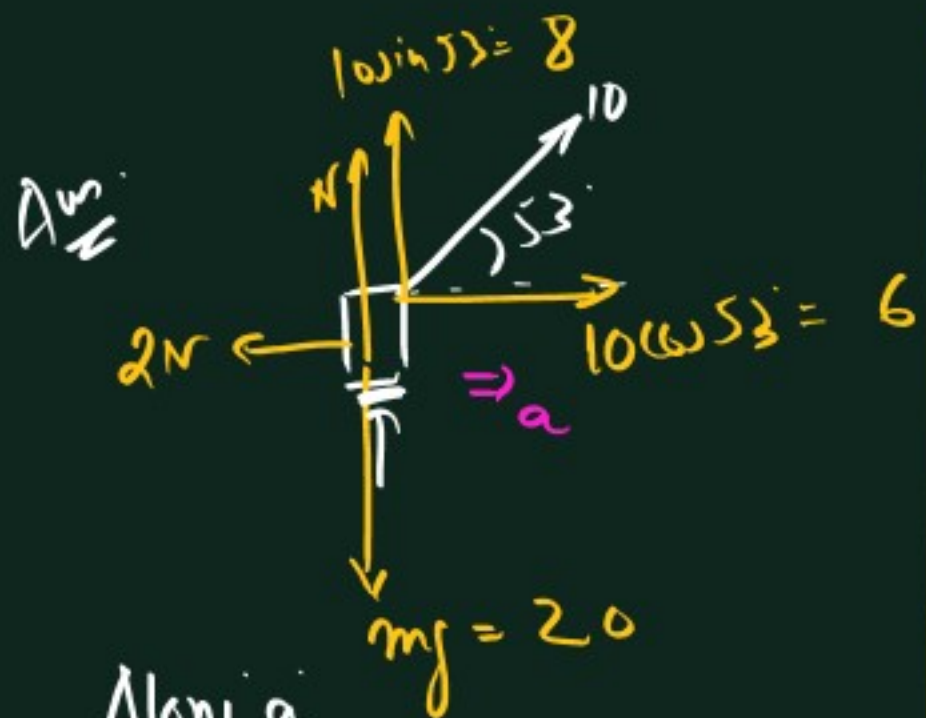
Q2



Find a & N

$$\cos 53^\circ = 3/5$$

$$\sin 53^\circ = 4/5$$



Along a

$$6 - 2 = 2a$$

$$\Rightarrow 4 = 2a$$

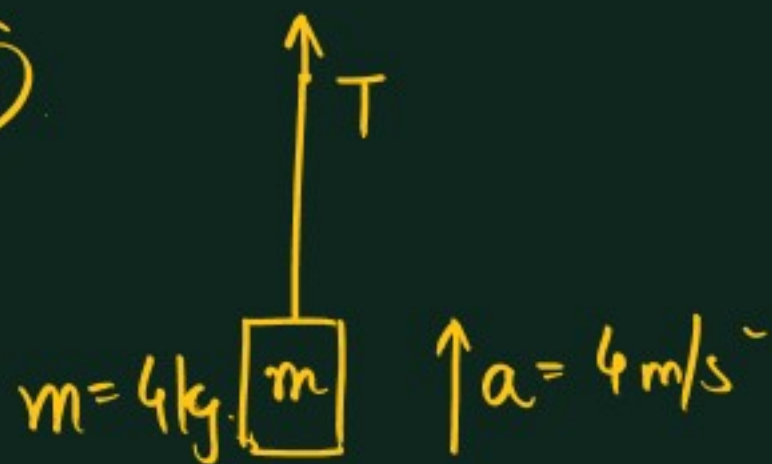
$$\Rightarrow \boxed{a = 2}$$

Perp to a $\Sigma F = 0$

$$\Rightarrow N + 8 = 20$$

$$\Rightarrow \boxed{N = 12}$$

Q3



Find T?



$\uparrow a$

$$T - mg = ma$$

$$\Rightarrow T = m(g + a)$$

$$= 4(10 + 4)$$

$$= \underline{\underline{56 \text{ N}}}$$

Q4



$$m = 2 \text{ kg}$$

Find N on box.

$$\Sigma F = ma$$

$$\Rightarrow mg - N = ma$$

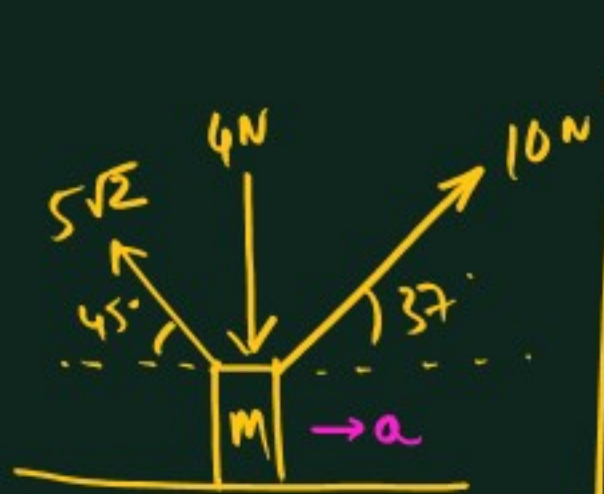
$$\Rightarrow N = m(g - a)$$

$$= \underline{\underline{16}}$$



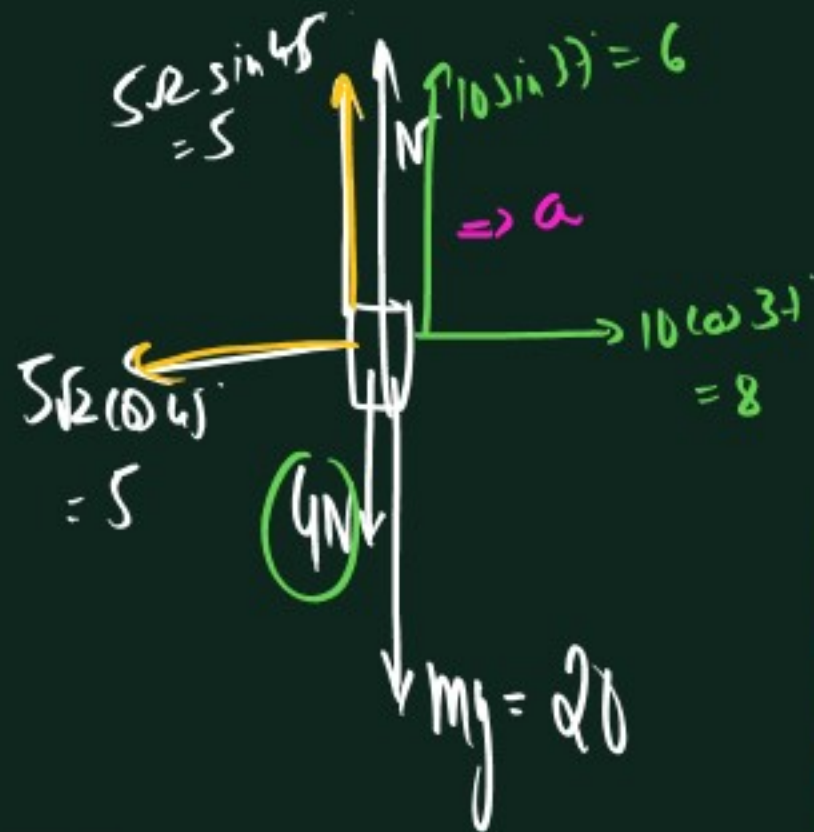
$$ma + mg = N$$

$\otimes$



2 kg

Find a & N



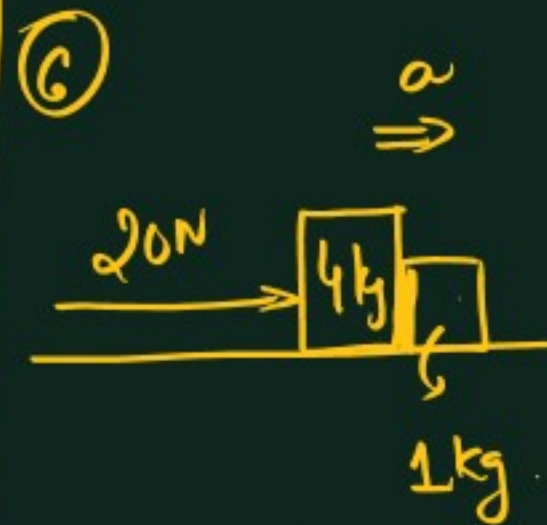
$$\Sigma F = ma$$

$$\Rightarrow 8 - 5 = 2a \Rightarrow \boxed{a = 3/2}$$

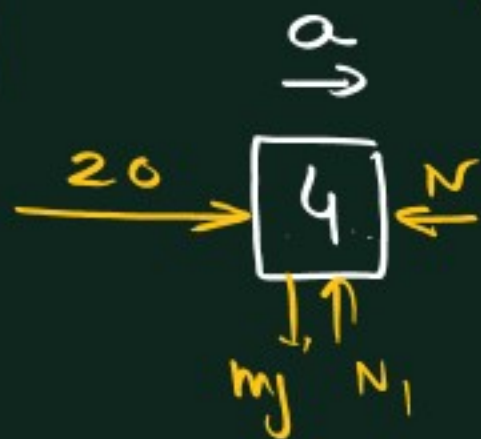
$$\textcircled{1} \Sigma F = 0$$

$$\Rightarrow N + 5 + 6 = 20 + 4$$

$$\Rightarrow \boxed{N = 13}$$



Find 1) a
2) Normal between blocks



$$\textcircled{4 \text{ kg}} \quad 20 - N = 4a \quad \text{--- (1)}$$

$$N = 1a \quad \text{--- (2)}$$

$$\text{Add } 20 = 5a \Rightarrow \boxed{a = 4}$$

$$\boxed{N = 4}$$

$$a = \frac{F}{m}$$

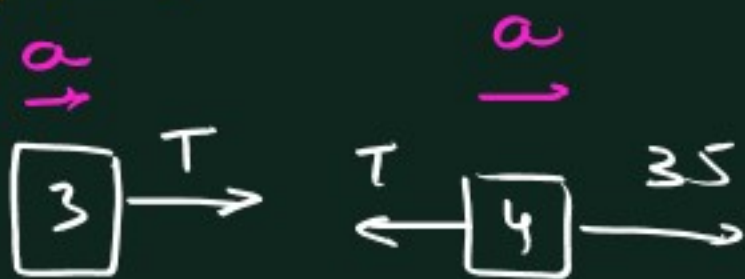
$$= \frac{20}{5} = \boxed{4}$$

$$N = 1 \times a$$

$$= \boxed{4}$$



Find a & Tension T .

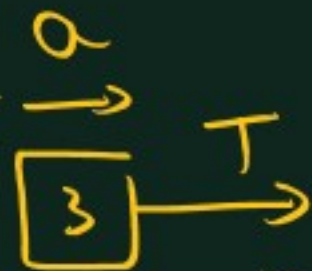


③ $T = 3a$ — ①

④ $35 - T = 4a$ — ②

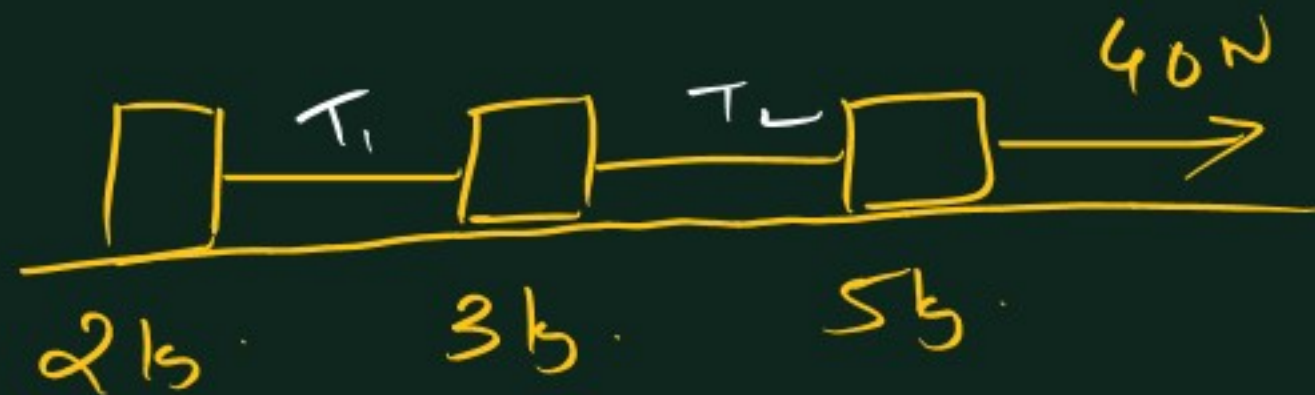
Solving $a = 5, T = 15$

Short way $a = \frac{35}{7} = 5$

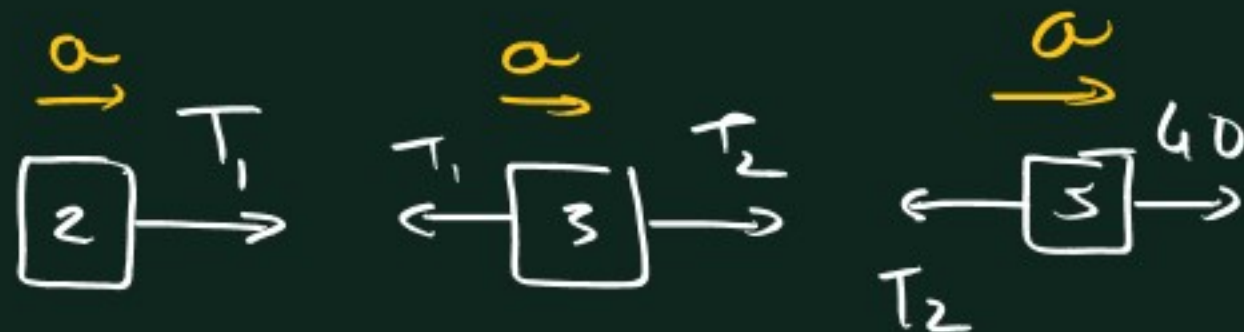


$T = 3a = 15$ Add

⑧



Find a, T_1, T_2 .



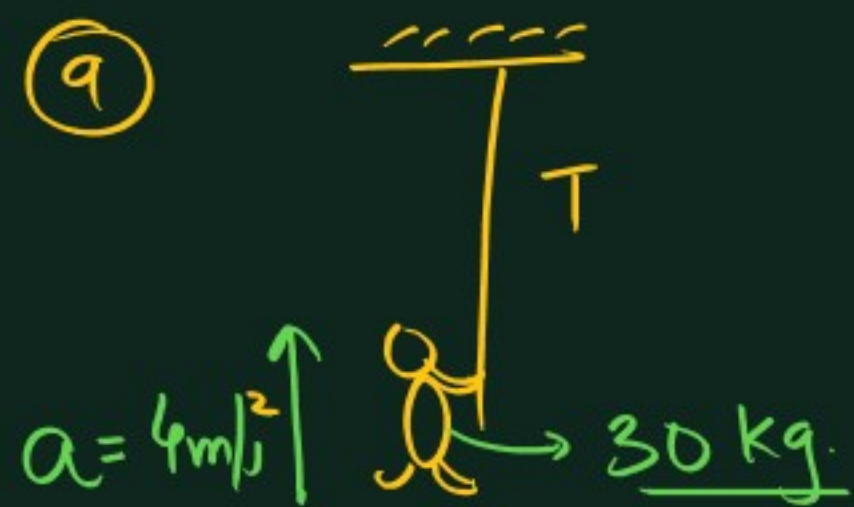
$T_1 = 2a$ — ①

$T_2 - T_1 = 3a$ — ②

$40 - T_2 = 5a$ — ③

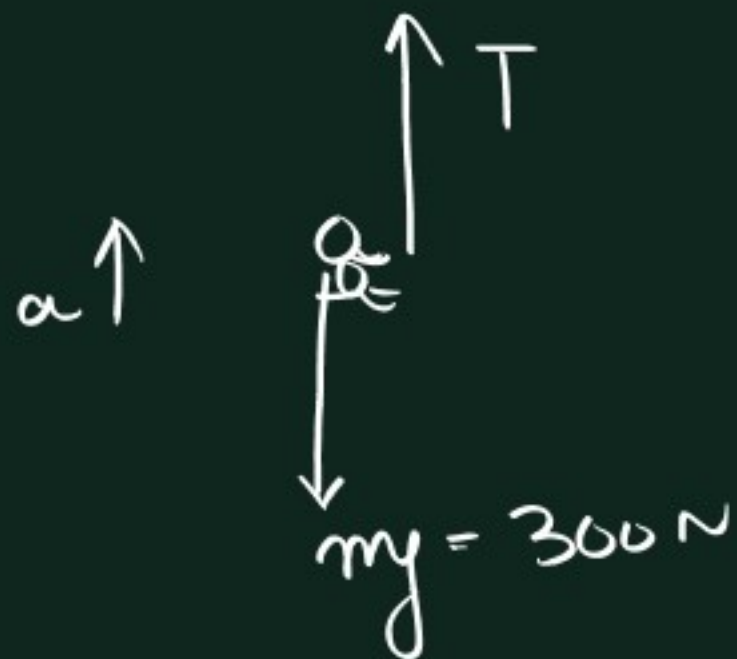
$40 = 10a \Rightarrow a = 4$, Put in ① & ②, $T_1 = 8, T_2 = 20$

Q



1) Find T in string

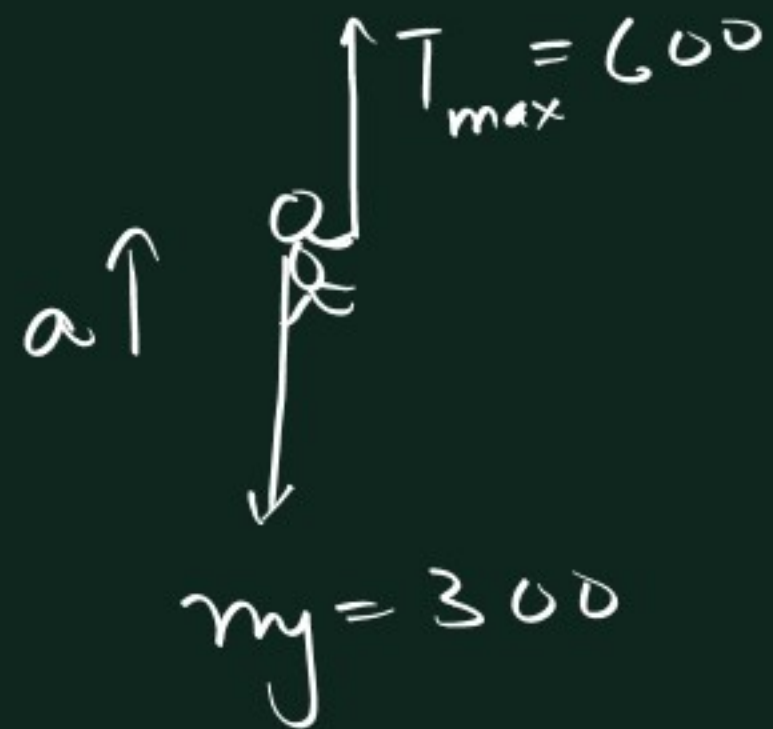
2) If breaking strength of string = 600 N, find max. possible acceleration of monkey



$$\Sigma F = ma$$

$$\Rightarrow T - 300 = 30 \times 4$$

$$\Rightarrow \boxed{T = 420 \text{ N}}$$



$$600 - 300 = ma$$

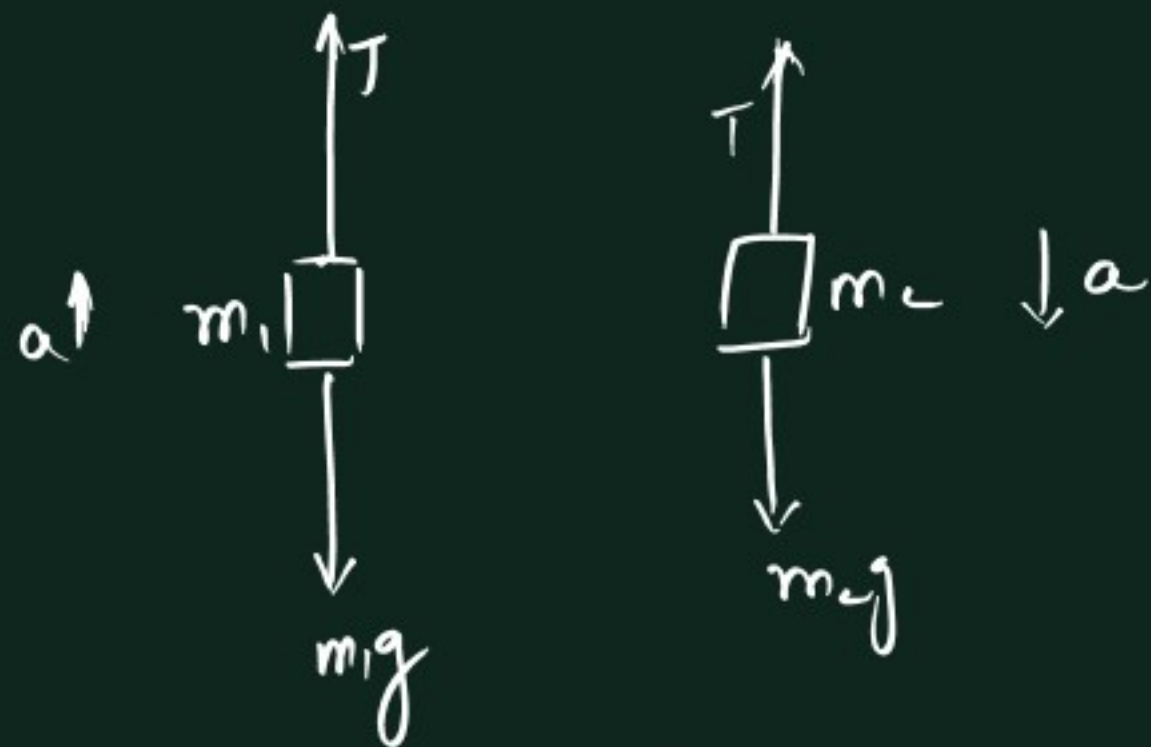
$$\Rightarrow a = \frac{300}{m} \\ = \underline{\underline{10 \text{ m/s}^2}}$$

Pulley block



Atwood machine

$m_2 > m_1$

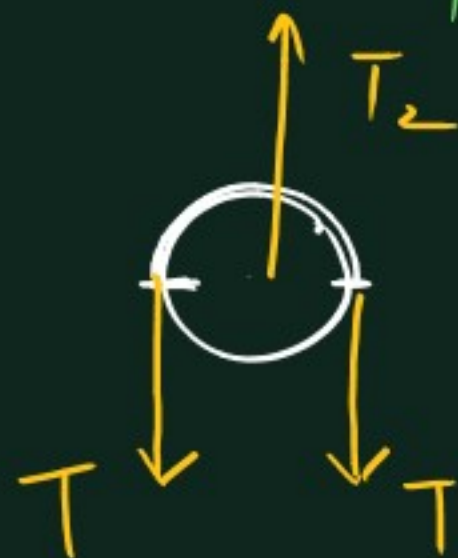


$$T - m_1g = m_1a \quad \text{--- (1)}$$

$$m_2g - T = m_2a \quad \text{--- (2)}$$

Solving $a = \frac{m_2g - m_1g}{m_1 + m_2}$, $T = \frac{2m_1m_2g}{m_1 + m_2}$

Find T_2 (in the string supporting massless pulley)

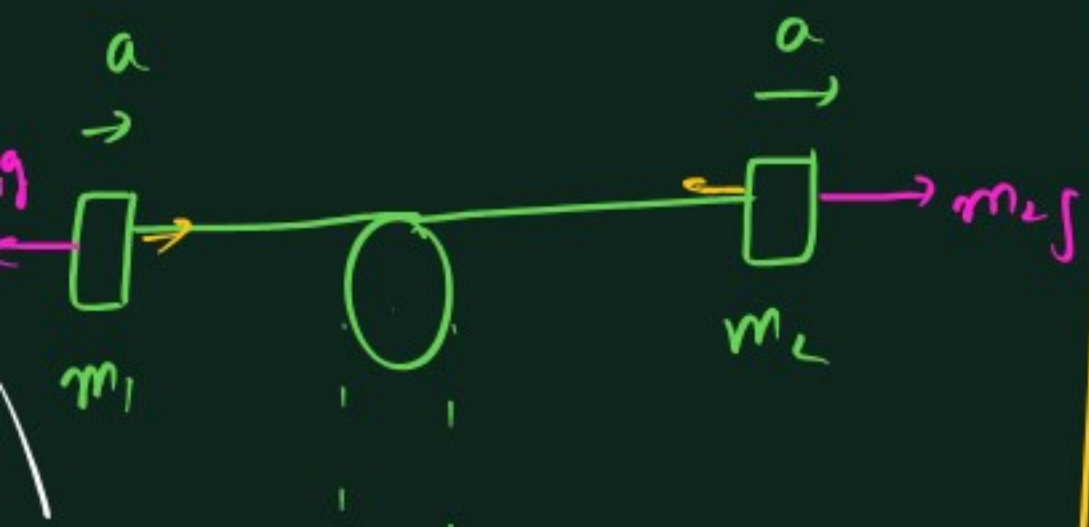


$$\sum F = 0$$

$$\Rightarrow T_2 = 2T$$

$$\sum F = ma$$

$$\Rightarrow \sum F = 0$$



$$a = \frac{\text{Net force}}{\text{total mass}} = \boxed{\frac{m_2g - m_1g}{m_1 + m_2}}$$

$a = \frac{\text{Net external pulling force along } a}{\text{total mass}}$

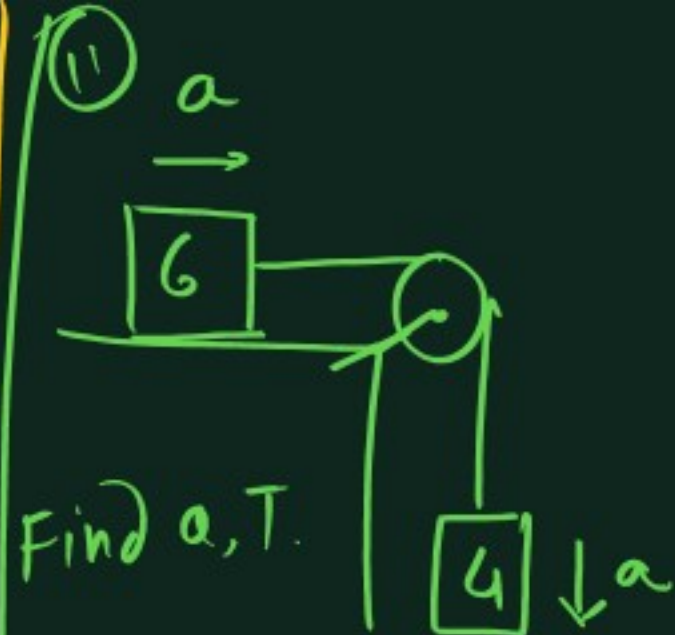
Q.10)



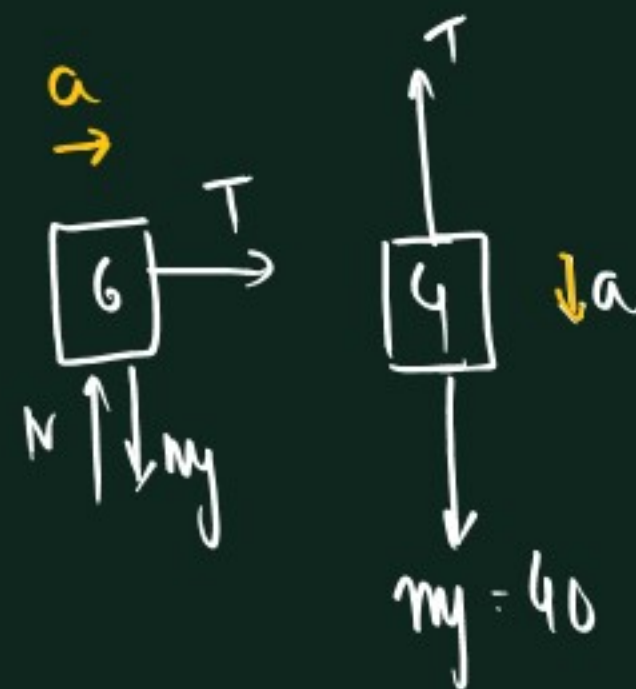
Find a & T

$$\boxed{a = 6}$$

$$\boxed{T = 16}$$



(4kg, 6kg)



(4kg)

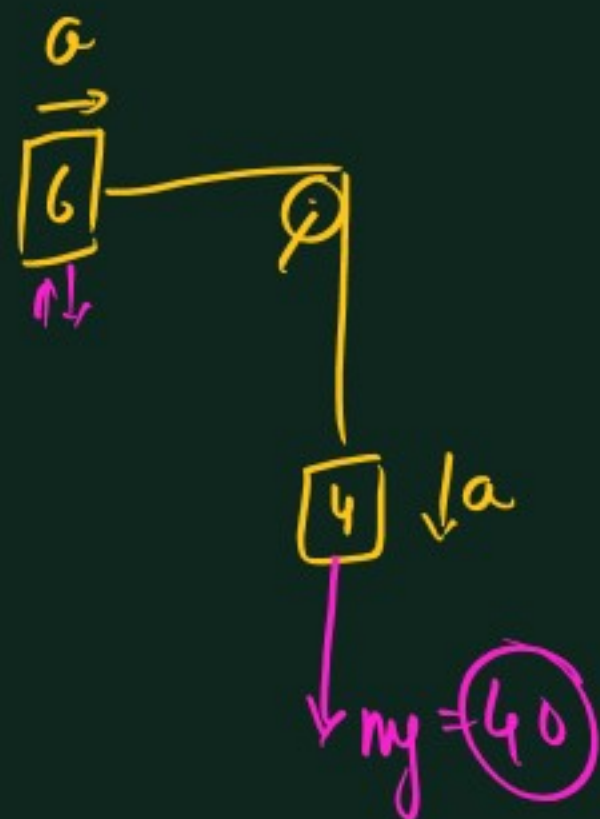
$$40 - T = 4a \quad \text{--- ①}$$

$$T = 6a \quad \text{--- ②}$$

Add $40 = 10a$

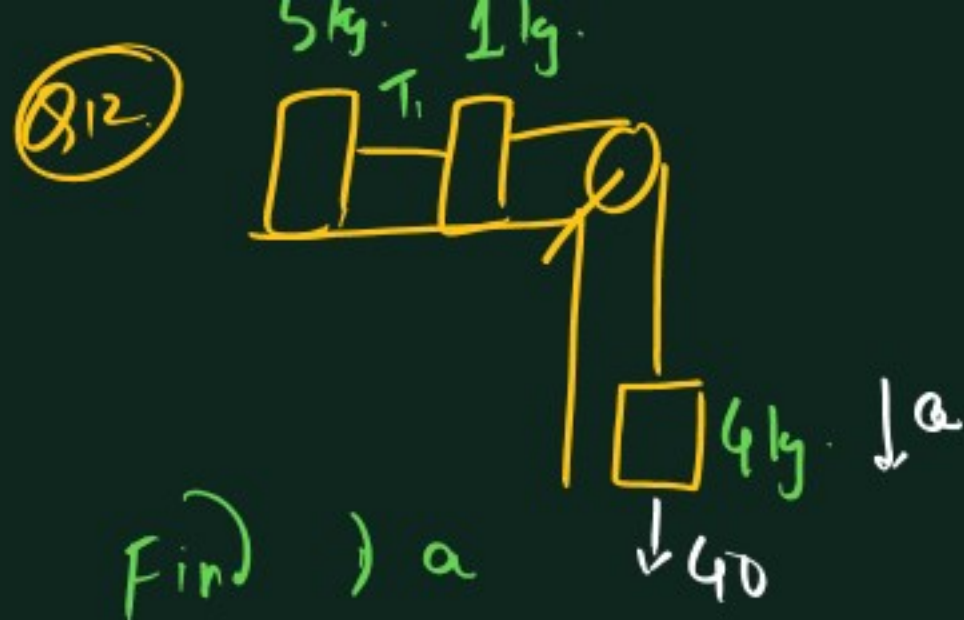
$$\Rightarrow \boxed{a = 4}$$

$$\boxed{T = 24}$$



$$a = \frac{\text{Net external pulling force}}{\text{Total } m} \text{ along } a$$

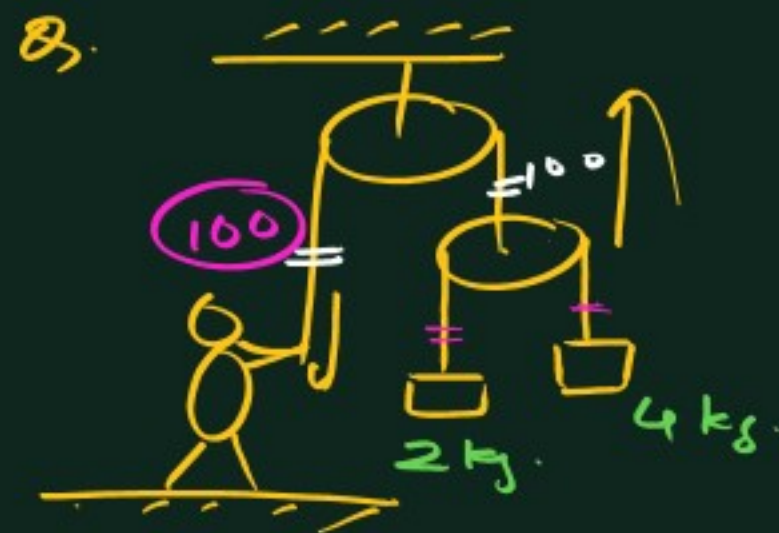
$$= \frac{40}{4+6} = \underline{\underline{4 \text{ m/s}^2}}$$



Find) a
e) T b/w 5 kg & 1 kg

$$a = \frac{\text{Fnet along } a}{\text{total mass}} = \frac{40}{5+1+4} = \underline{\underline{4}}$$

$$T_1 = 5a = 20 \text{ N}$$



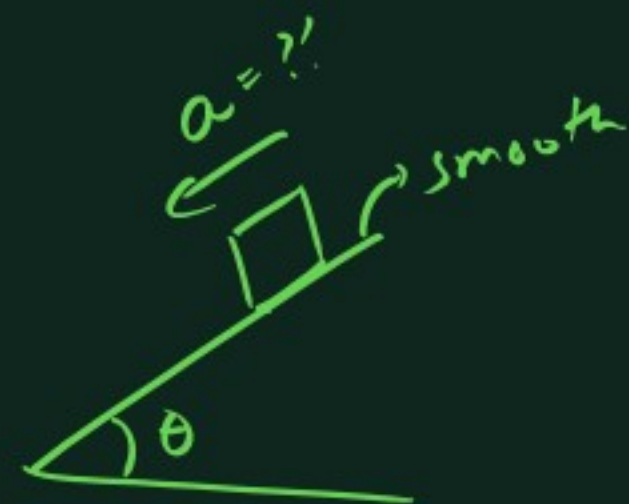
$\Sigma F = 0$
(massless)

$\Rightarrow 2T = 100$
 $\Rightarrow \underline{\underline{T = 50}}$

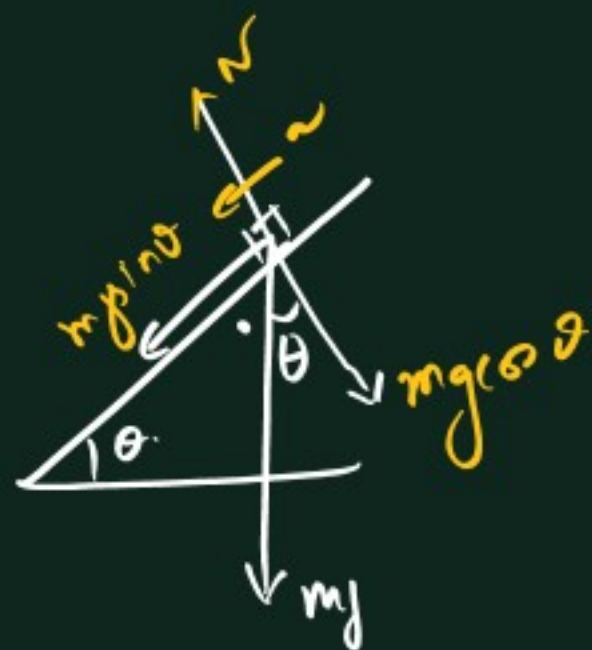
$50 - 20 = 2a_1$
 $\Rightarrow \underline{\underline{a_1 = 15}}$

$50 - 40 = 4a_2$
 $\Rightarrow \underline{\underline{a_2 = 2.5}}$

Q13.

Find a & Normal on box!

- A) g B) $g \sin \theta$
 C) $g \cos \theta$ D) $g \tan \theta$

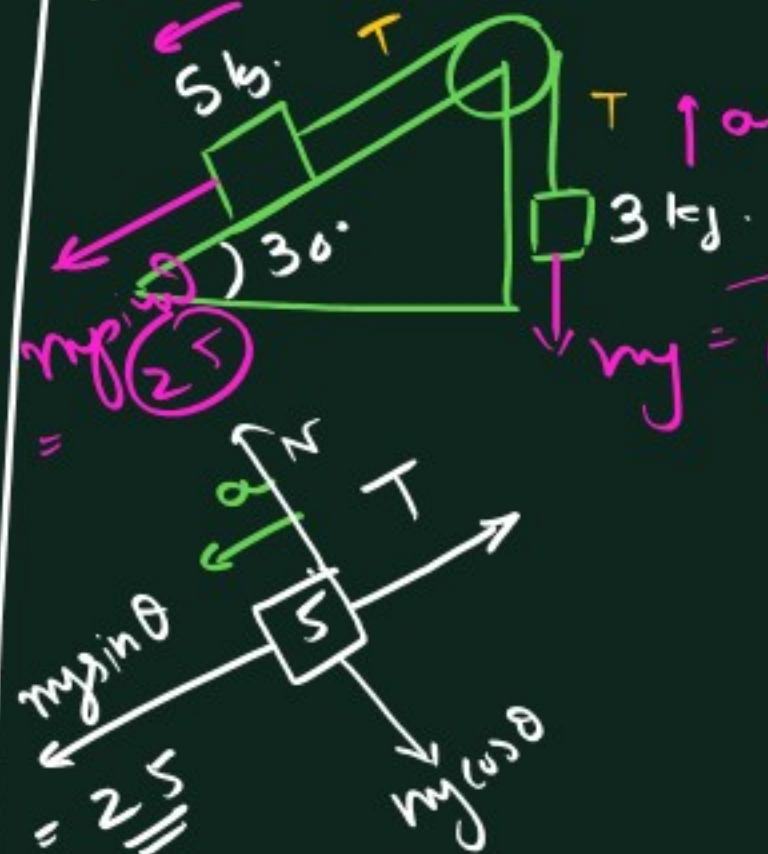


Along a : $mg \sin \theta = ma$
 $\Rightarrow a = g \sin \theta$

(1)

$N = mg \cos \theta$

Q14.

Find 1) a
2) T

(5 kg) $\Sigma F = ma \Rightarrow 25 - T = 5a \quad \text{--- (1)}$

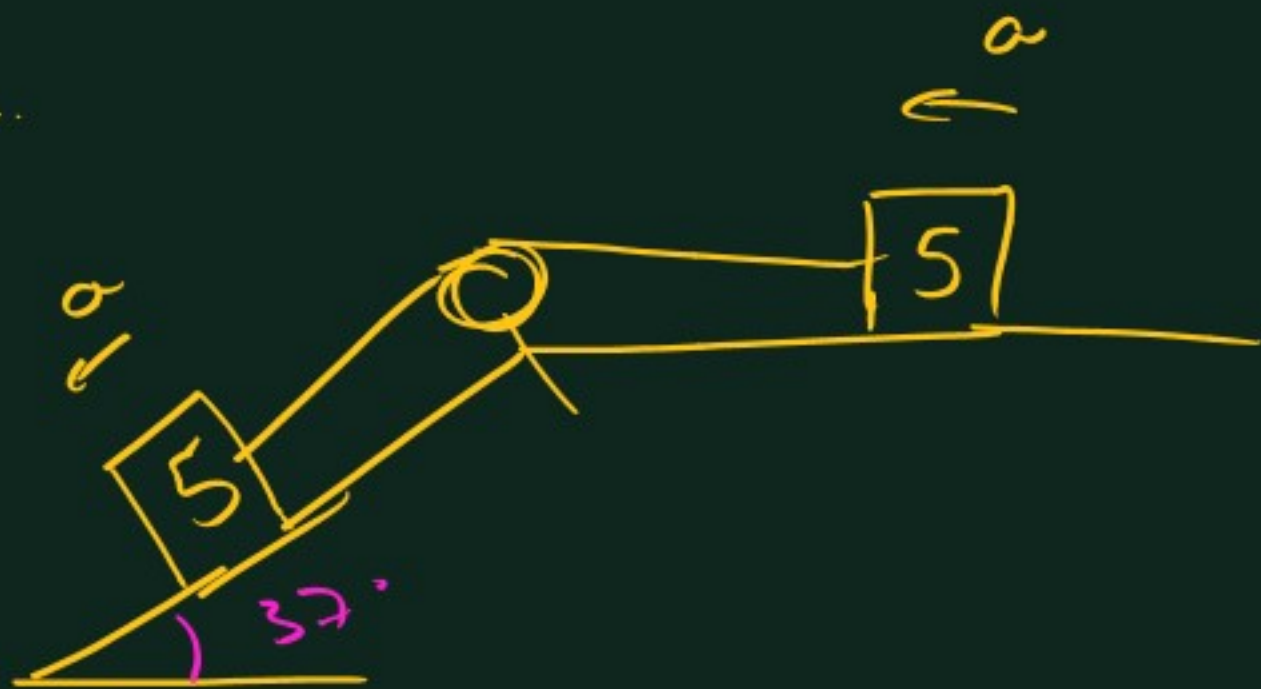
(3 kg) $\Sigma F = ma \Rightarrow T - 30 = 3a \quad \text{--- (2)}$

Solving $a = -5/8$

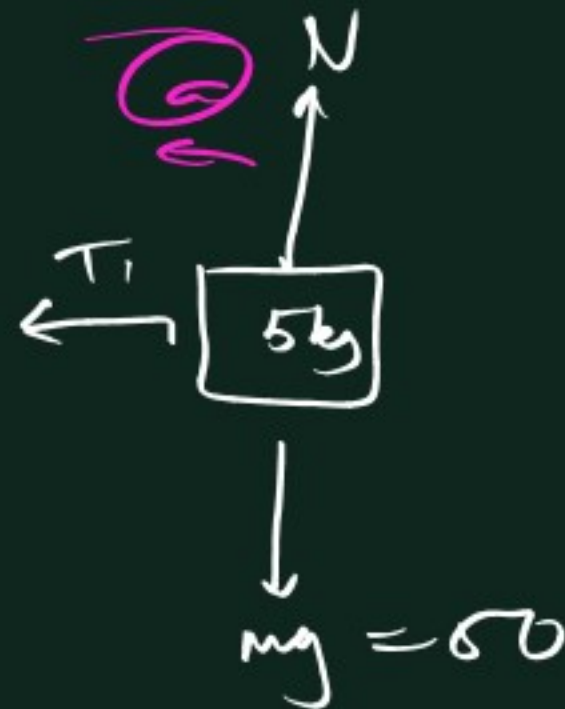
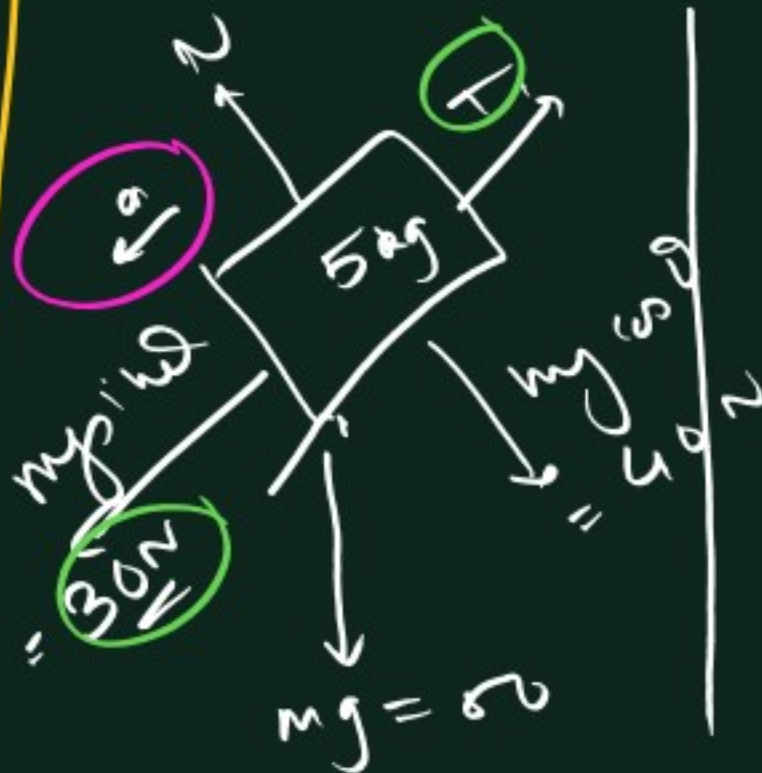
$T = 30 + 3(-5/8) = \frac{225}{8}$



Q15



Both blocks of 5 kg mass
Find a , T .



Incline $30 - T = 5a$ — (1)

Horizontal $T = 5a$ — (2)

Add $30 = 10a \Rightarrow a = 3$

$T = 15$

$\downarrow g$



$$\text{Reading} = \frac{\text{Normal}}{g}$$

A) 0
C) ∞

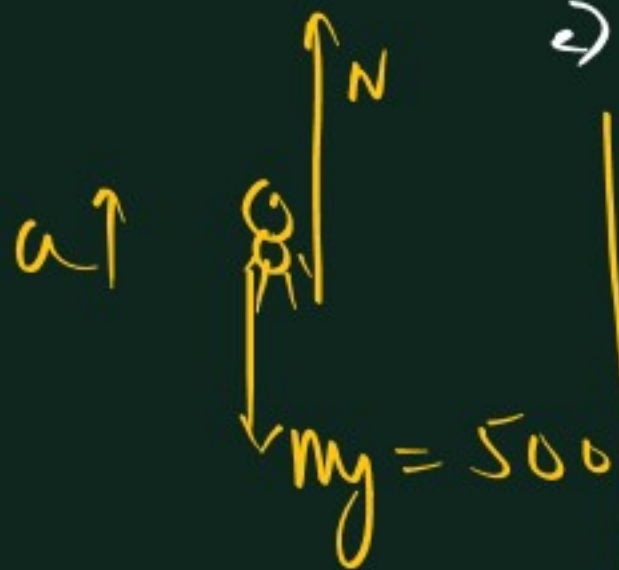
B) 50 kg.
D) 5 kg.

Q



$\uparrow a = 2 \text{ m/s}^2$

Find reading of machine
1) in Newton
2) in kg.



$$\Sigma F = ma$$

$$\Rightarrow N - 500 = 50 \times 2$$

$$\Rightarrow \boxed{N = 600}$$

$$\begin{aligned} \text{Reading in Newtons} &= \underline{600 \text{ N}} \\ \text{" " kg.} &= \frac{600}{10} = \underline{60 \text{ kg}} \end{aligned}$$



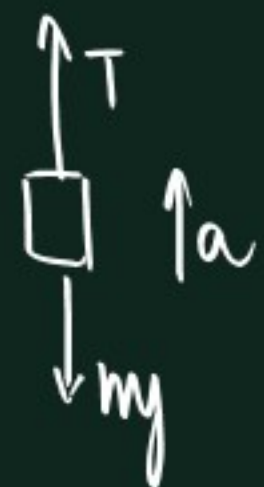
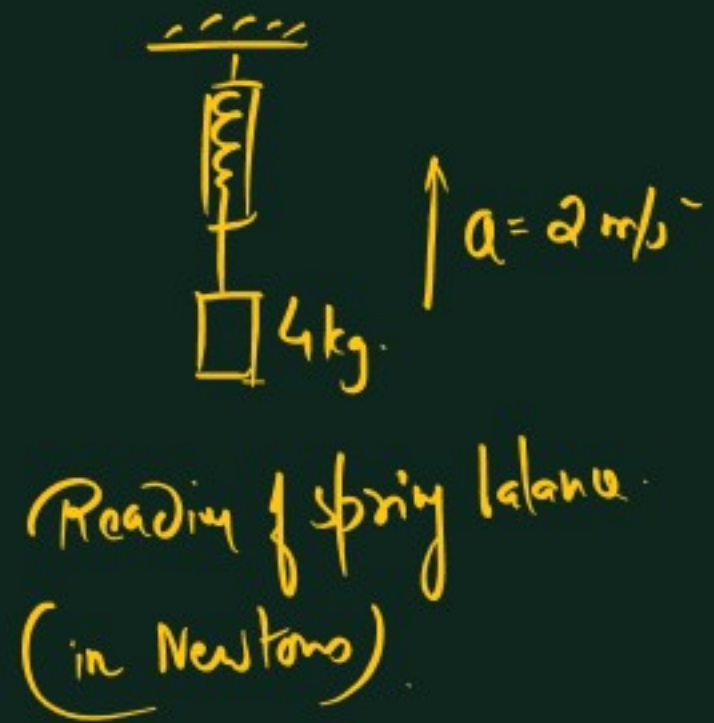
$$\Sigma F = ma$$

$$\Rightarrow mg - N = m \textcircled{a} \rightarrow a=g$$

$$m\cancel{g} - N = m\cancel{g}$$

$$\Rightarrow \boxed{N=0}$$

Q1.



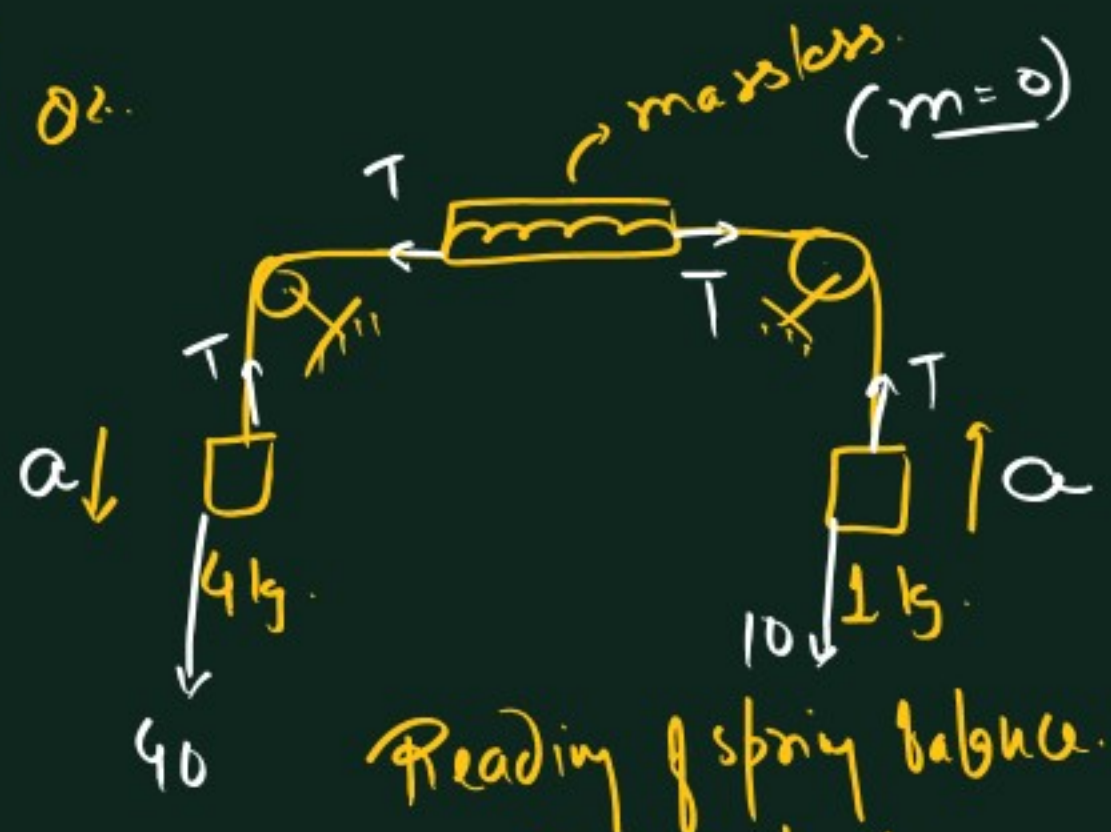
$$T - mg = ma$$

$$\Rightarrow T = mg + ma$$

$$= \underline{\underline{48}}$$

80 ps

Q2.



$\Rightarrow$ As spring massless $\Rightarrow \Sigma F = 0$

$$\Rightarrow T_1 = T_2 = T$$

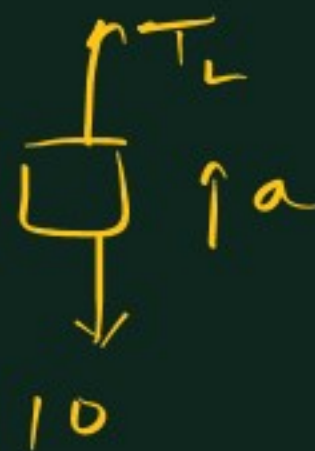
$$40 - T = 4a \quad \text{--- (1)}$$

$$T - 10 = 1a \quad \text{--- (2)}$$

Solving, $\underline{\underline{T = 16 \text{ N}}}$

$$T = \frac{2m_1 m_2}{m_1 + m_2}$$

$$= 16 \text{ N.}$$



$$40 - T_1 = 4a \quad \text{--- (1)}$$

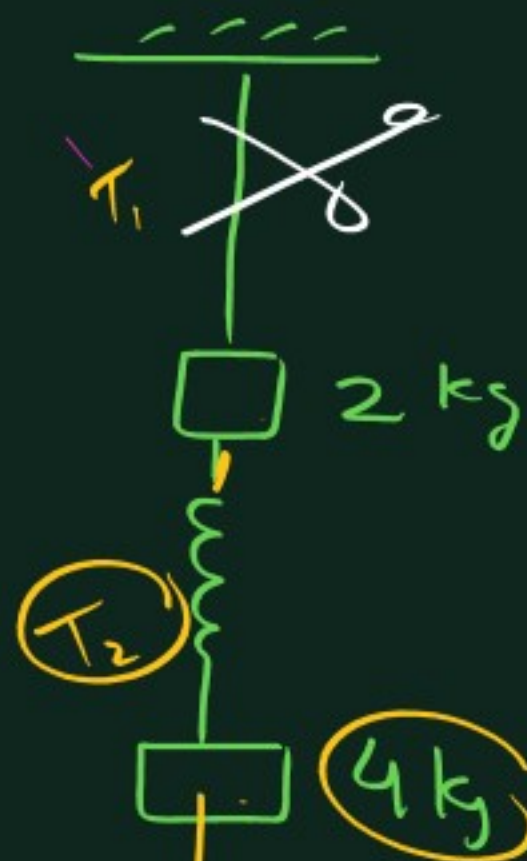
$$T_1 - T_2 = ma \quad \text{--- (2)}$$

$$T_2 - 10 = 1a \quad \text{--- (3)}$$

Solve for T_1, T_2, a .

80 ps (string)

a.

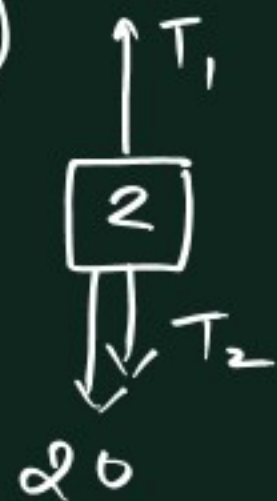


1) System at rest

Find T_1 , T_2

2) If rope (string) is cut, find acceleration of blocks just after cutting.

1)



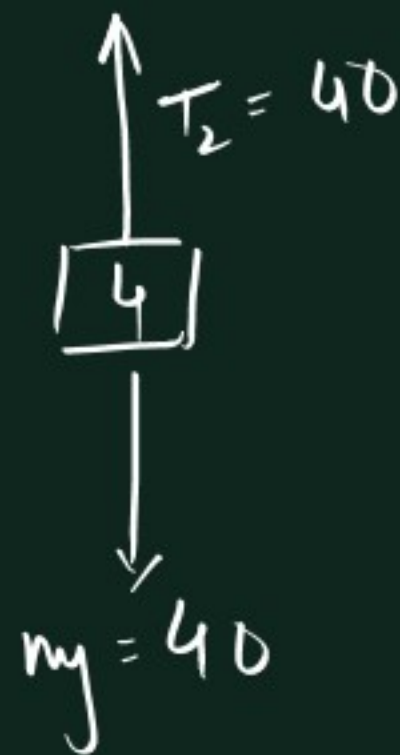
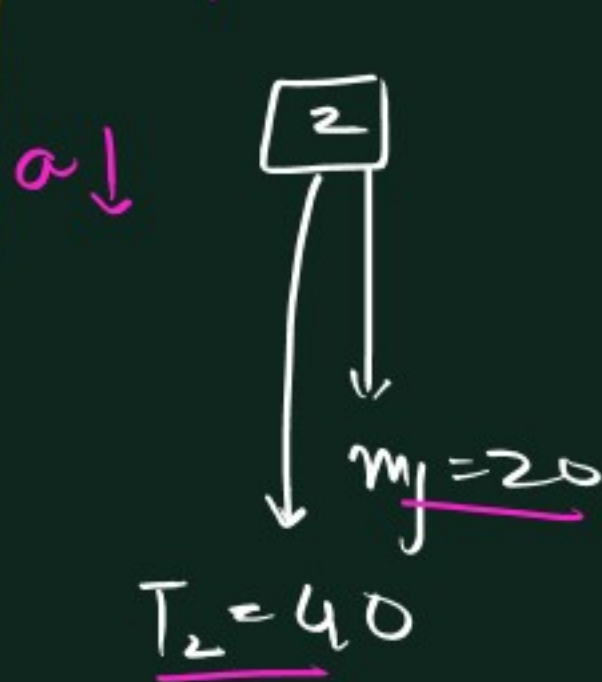
$$T_1 = T_2 + 20 \quad \text{--- (1)}$$

$$T_2 = 40 \quad \text{--- (2)}$$

solving

$$\boxed{\begin{matrix} T_1 = 60 \\ T_2 = 40 \end{matrix}}$$

2) T_{spring} after cutting
spring force stays 40N



(2 kg)

$$\Sigma F = ma$$

$$\Rightarrow 40 + 20 = 2a$$

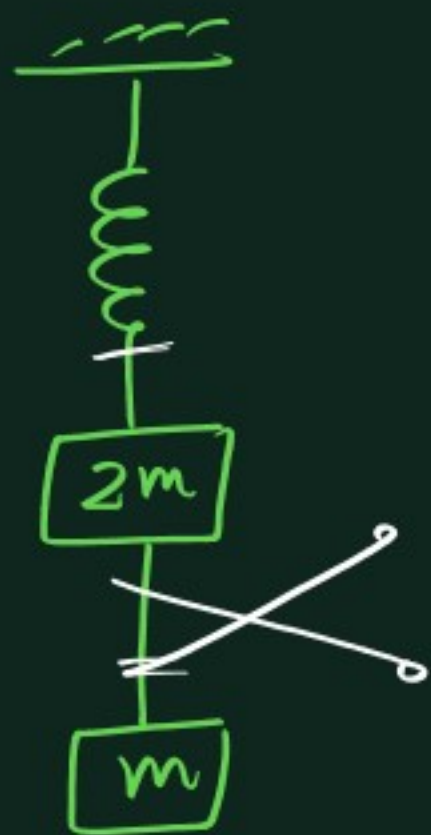
$$\Rightarrow a = 30 \text{ m/s}^2 (\downarrow)$$

(4 kg)

$$\Sigma F = 0 \quad (T_2 = m_2 g)$$

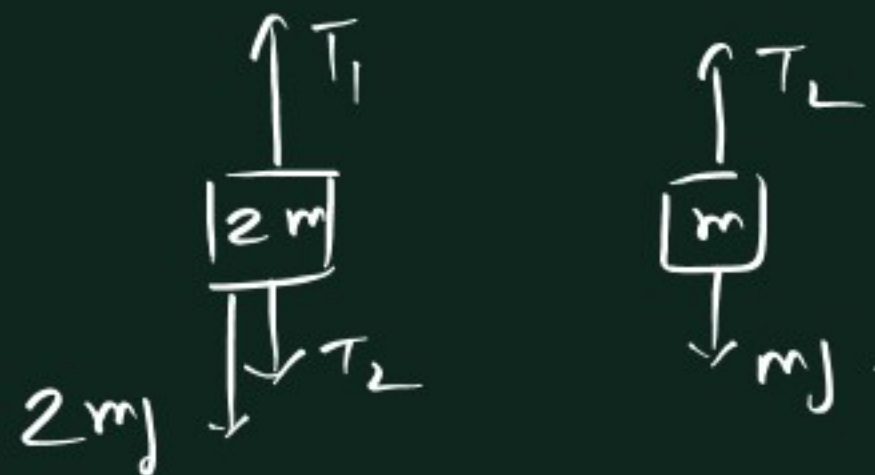
$$\Rightarrow \underline{\underline{a = 0}}$$

Q



Acceleration of $2m$
 & m just after
 cutting.

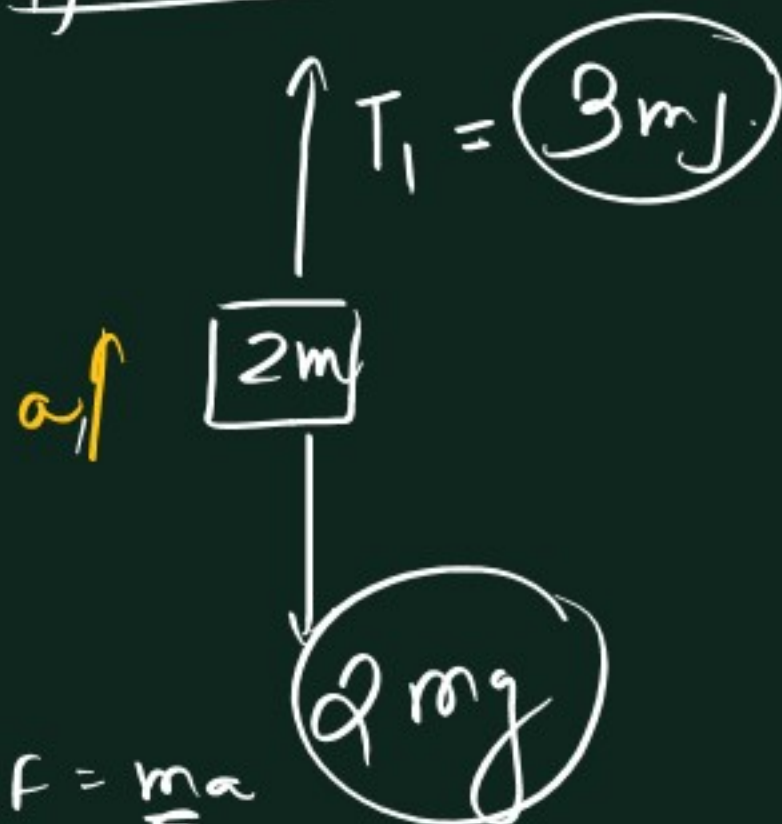
Before cutting.



m $T_2 = mg$

$2m$ $T_1 = T_2 + 2mg$
 $\Rightarrow T_1 = 3mg$

After cut



$\Sigma F = ma$

$3mg - 2mg = (2m)a_1$

$\Rightarrow a_1 = g/2 (\uparrow)$

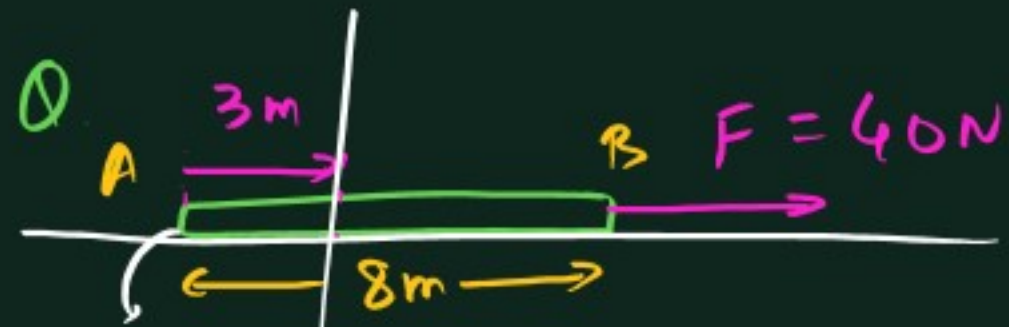


$\Sigma F = ma$

$\Rightarrow mg = ma$

$\Rightarrow a = g (\downarrow)$

More Problems



Rope mass = 10kg.

Find 1) a of rope

2) T at 3m from 'a'

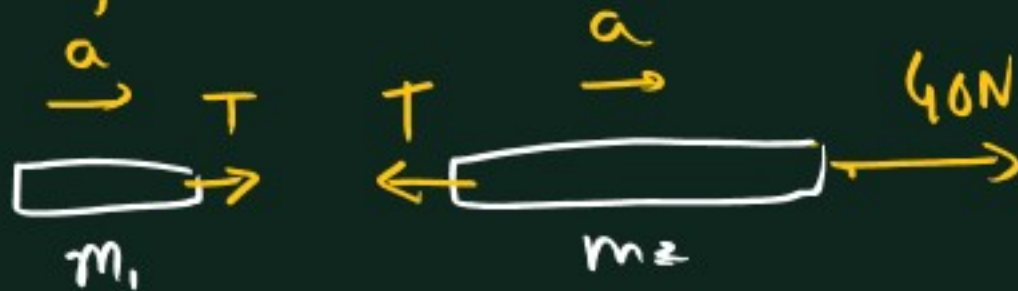
** If string has mass, T different at different points.



$$\Sigma F = ma$$

$$\Rightarrow 40 = 10a \Rightarrow \underline{a = 4}$$

* If T at a point in string with mass required, cut at that point & draw FBD of 2 parts



$$m_1 = \left(\frac{l_1}{L}\right) \times m = \frac{3}{8} \times 10 = \underline{\underline{\frac{30}{8} \text{ kg}}}$$

$$m_2 = \frac{5}{8} \times 10 = \underline{\underline{\frac{50}{8} \text{ kg}}}$$

m_1

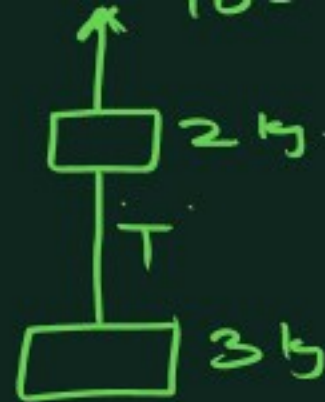
$$T = m_1 a$$

$\hookrightarrow 4$

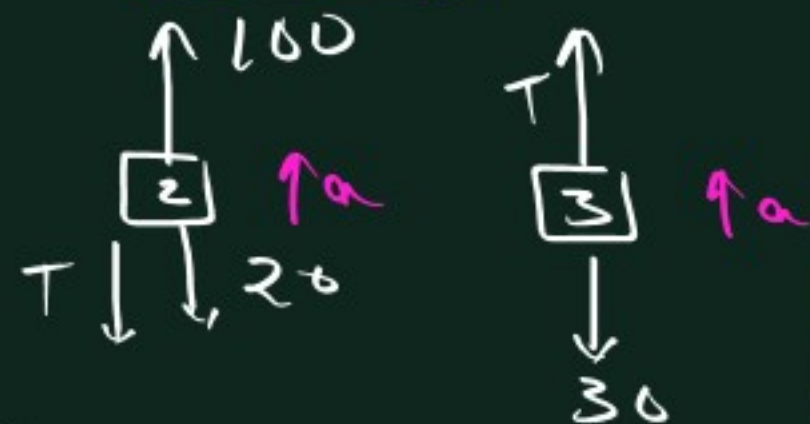
$$\Rightarrow T = \frac{30}{8} \times 4$$

$$\Rightarrow \boxed{T = 15\text{N}}$$

Q1.



Find T in string.

60 N

$$100 - T - 20 = 2a \quad \text{--- (1)}$$

$$T - 30 = 3a \quad \text{--- (2)}$$

$$a = 10$$

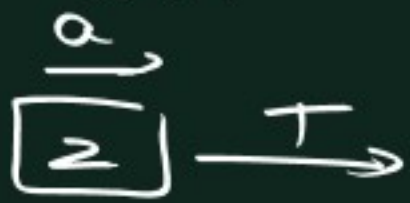
$$T = 60$$

Q2.

**20/3**

Find T in string

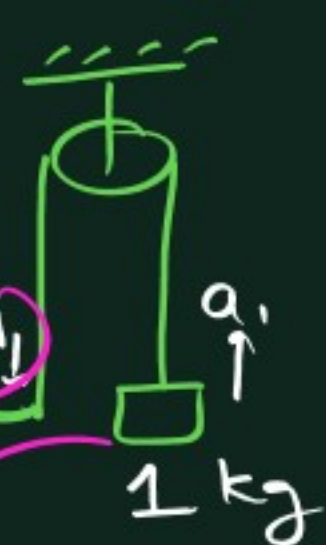
$$a = \frac{10}{2+1} = 10/3$$



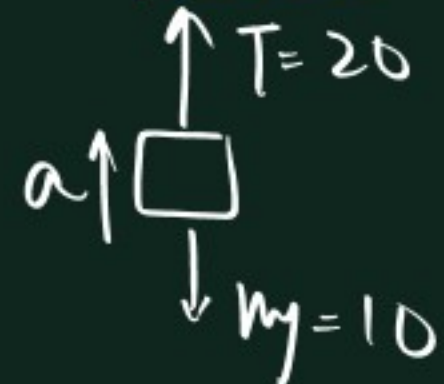
$$T = ma$$

$$\Rightarrow T = 2 \times \frac{10}{3} = \frac{20}{3}$$

Q3.



man applies 20 N

Find $a_1 : a_2$ **3:1** **$T = 20 \text{ N}$** 

$$\Sigma F = ma$$

$$\Rightarrow 20 - 10 = 1a$$

$$\Rightarrow \boxed{a_1 = 10}$$

Case II $a = \frac{20 - 10}{2+1} = \frac{10}{3}$